

## TREATMENT AND EMERGENCY RECOMMENDATIONS FOR ISOVALERIC ACIDEMIA

PATIENT: \_\_\_\_\_ MRN: \_\_\_\_\_ DOB: \_\_\_\_\_

**This patient has ISOVALERIC ACIDEMIA and is at risk for serious complications from this condition.**

### **Please see last section of this letter for Emergency Treatment during Decompensation**

Definition: Isovaleric Acidemia is a rare inherited metabolic disorder of Leucine metabolism. Isovaleryl CoA is an organic acid metabolite that occurs early in the catabolism of leucine. In the absence of isovaleryl-CoA dehydrogenase the organic acid IVA is neurotoxic and can cause lethargy, hypotonia, feeding problems, vomiting, coma, cerebral edema, and even death.

Signs of Decompensation: Lethargy, hypotonia, feeding problems, vomiting, coma, and/or cerebral edema.

#### Treatments:

- A. **Maintenance Treatment:** is successful if the Protocol is followed.
  - Diet:** Diet/Metabolic Formula with minimal leucine.
  - L-Carnitine 100mg/kg/day
  - Glycine 250mg/kg/day
  - Aggressive treatment during illness to prevent metabolic decompensation
- B. **Treatments during an illness** (colds, flu, ear infections, etc):
  - Prevent catabolism by:**
    - i. Maintaining liquids orally that contain glucose
    - ii. Maintaining L-Carnitine and Glycine intake
    - iii. If unable to take oral liquids then escalate to the EMERGENCY PROTOCOL
- C. **Treatment during Decompensation: Emergency Protocol**
  - Vomiting, Lethargy, Ketoacidosis, etc (all or part of these symptoms)*
  - 1. Restore an anabolic state

- i. IV Glucose (D10/ ½ NS + KCL)
  - ii. IV L-Carnitine 200mg/kg over 2 hours (compatible with D10). Then L-Carnitine 100mg/kg/day in continuous infusion or q 6hrs
  - iii. Bicarbonate 1.0 meq/kg diluted in IV fluids PRN
  - iv. Labs- DO NOT WAIT for labs to start therapy (items A and B).  
BMP, NH3, Lactate, CBC
  - v. Treat underlying illness (infection, etc)
2. Observe response
  - i. Monitor urine ketones for effectiveness of therapy. Switch to oral medications L-carnitine and glycine during convalescence

**Notify the metabolic specialist (below) immediately. After business hours calling this number allows you to speak with the geneticist on call:**

**Kentucky Children's Hospital**  
**Division of Genetics and Metabolism**  
R. Stephen Amato, MD, PhD  
859-323-0396  
Kentucky Clinic – J420  
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**(After Hours: Please call 1-800-888-5533)**